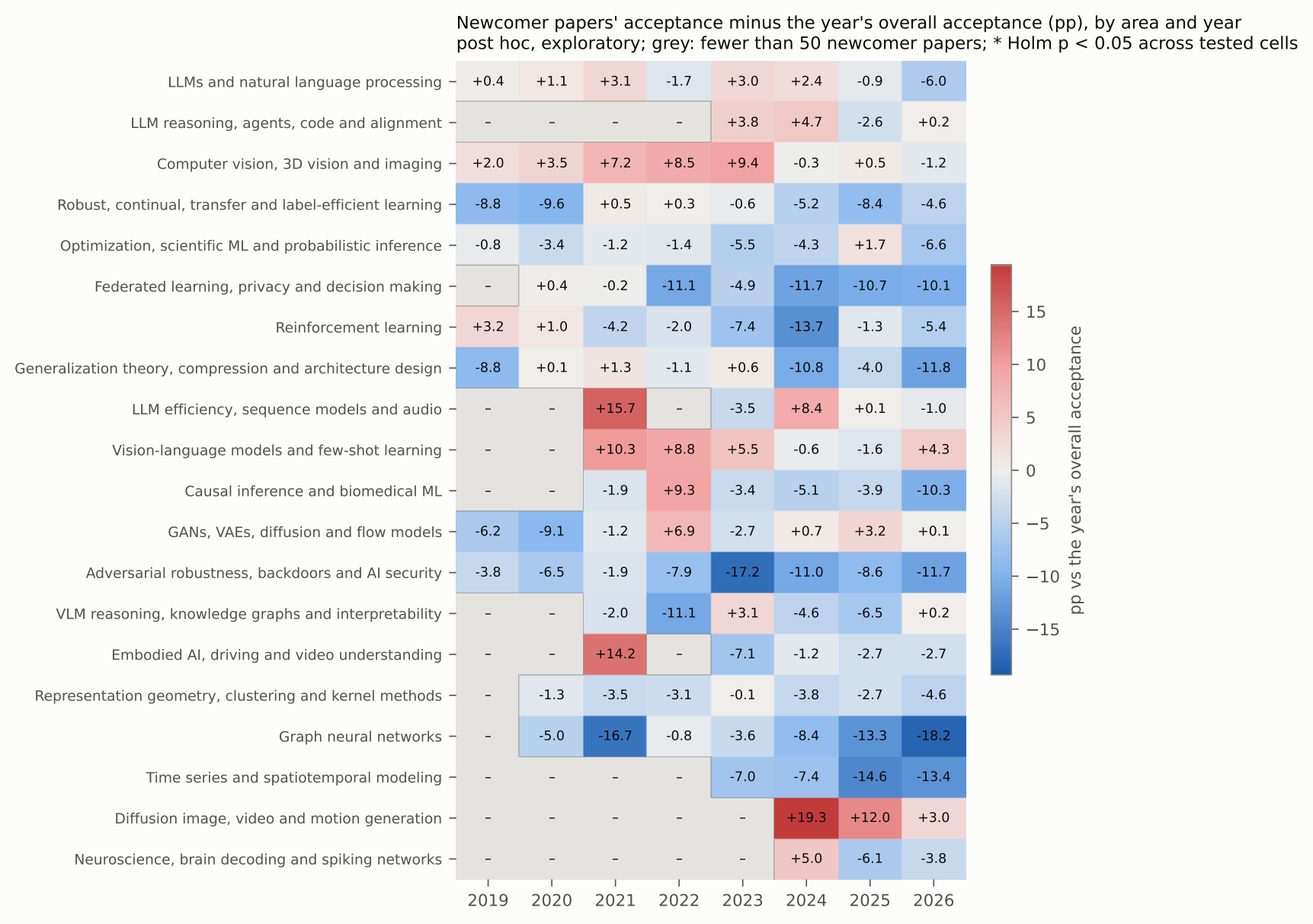
Newcomer papers' acceptance minus the year's overall acceptance (pp), by area and year
post hoc, exploratory; grey: fewer than 50 newcomer papers; * Holm p < 0.05 across tested cells
LLMs and natural language processing +0.4 +1.1 +3.1 -1.7 +3.0 +2.4 -0.9 -6.0
LLM reasoning, agents, code and alignment – – – – +3.8 +4.7 -2.6 +0.2
Computer vision, 3D vision and imaging +2.0 +3.5 +7.2 +8.5 +9.4 -0.3 +0.5 -1.2
Robust, continual, transfer and label-efficient learning -8.8 -9.6 +0.5 +0.3 -0.6 -5.2 -8.4 -4.6
Optimization, scientific ML and probabilistic inference -0.8 -3.4 -1.2 -1.4 -5.5 -4.3 +1.7 -6.6
Federated learning, privacy and decision making – +0.4 -0.2 -11.1 -4.9 -11.7 -10.7 -10.1
Reinforcement learning +3.2 +1.0 -4.2 -2.0 -7.4 -13.7 -1.3 -5.4
Generalization theory, compression and architecture design -8.8 +0.1 +1.3 -1.1 +0.6 -10.8 -4.0 -11.8
LLM efficiency, sequence models and audio – – +15.7 – -3.5 +8.4 +0.1 -1.0
Vision-language models and few-shot learning – – +10.3 +8.8 +5.5 -0.6 -1.6 +4.3
Causal inference and biomedical ML – – -1.9 +9.3 -3.4 -5.1 -3.9 -10.3
GANs, VAEs, diffusion and flow models -6.2 -9.1 -1.2 +6.9 -2.7 +0.7 +3.2 +0.1
Adversarial robustness, backdoors and AI security -3.8 -6.5 -1.9 -7.9 -17.2 -11.0 -8.6 -11.7
VLM reasoning, knowledge graphs and interpretability – – -2.0 -11.1 +3.1 -4.6 -6.5 +0.2
Embodied AI, driving and video understanding – – +14.2 – -7.1 -1.2 -2.7 -2.7
Representation geometry, clustering and kernel methods – -1.3 -3.5 -3.1 -0.1 -3.8 -2.7 -4.6
Graph neural networks – -5.0 -16.7 -0.8 -3.6 -8.4 -13.3 -18.2
Time series and spatiotemporal modeling – – – – -7.0 -7.4 -14.6 -13.4
Diffusion image, video and motion generation – – – – – +19.3 +12.0 +3.0
Neuroscience, brain decoding and spiking networks – – – – – +5.0 -6.1 -3.8
2019 2020 2021 2022 2023 2024 2025 2026
15 10 5 0 −5 −10 −15
pp vs the year's overall acceptance